MD NAHID HASAN

AI/ML RESEARCHER • MACHINE LEARNING ENGINEER

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LinkedIn • GitHub • Portfolio

PROFILE

Computer Science PhD student and AI/ML researcher with experience spanning computer vision, reinforcement learning, scientific computing, and applied machine learning. Published work in *IEEE Access* and experienced in developing reproducible ML pipelines using Python, PyTorch, TensorFlow, Scikit-learn, and PySCF.

EDUCATION

PhD in Computer Science — 2026 – Present
University of North Carolina at Charlotte, Charlotte, NC
Research: Software Engineering for AI / LLMs

MSCS in Computer Science — Aug 2024 – May 2026
Miami University, Oxford, OH
GPA: 4.00/4.00

RESEARCH & INDUSTRY EXPERIENCE

Machine Learning / Computer Vision Intern — Summer 2026
Innovision LLC — Remote

- Developed computer vision pipelines for objectively tracking burn-scar healing using 3D facial scans from real burn patients.
- Built and evaluated deep learning segmentation approaches for 3D medical imaging, benchmarking models for a specialized healthcare application.
- Developed a browser-based visualization and correction tool for SAM2 segmentation polygons, scan overlays, alignment/feature JSON, and 3D `.ply` reconstructions; deployed via Render and GitHub.

Research Intern – Quantum Computing & Molecular Simulation — May 2025 – Aug 2025
University of Dayton Research Institute (UDRI) — Dayton, OH

- Investigated electron dynamics and charge transfer in DNA-based molecular systems using computational quantum methods.
- Applied RT-TDDFT and DFT to simulate molecular excitation and energy redistribution across nucleobases using PySCF-based Python workflows.
- Built reproducible analysis pipelines with NumPy and Matplotlib for molecular simulation and visualization.

Graduate Research & Teaching Assistant — Aug 2024 – May 2026
Miami University, Oxford, OH

- Developed **QuikDel**, a hierarchical multi-agent Q-Learning system for urban delivery, reducing travel distance by up to 25% while simulating 13,866+ requests/hour.
- Designed experiments and performance evaluations for multi-agent reinforcement learning; published results in *IEEE Access*.
- Mentored 60+ students in Machine Learning, Artificial Intelligence, and Systems Engineering courses and projects.

SELECTED PUBLICATIONS

M.N. Hasan et al., **"QuikDel: A Scalable Hierarchical Multi-agent System for Urban Delivery Using Q-Learning,"** *IEEE Access*, 2026.

Additional manuscripts under review at ICAPS 2026 and IEEE Quantum Software (QSW) 2026.

SELECTED ML PROJECTS

Deep Learning-Based Road Skeletonization — *PyTorch, U-Net*

- Developed a U-Net segmentation model for extracting one-pixel-wide road skeletons from noisy satellite imagery; achieved Dice = 0.87 and IoU = 0.79 on OpenStreetMap data.

Neural Processing of Emotional Musical Stimuli — *Python, SVM, PCA*

- Applied PCA + SVM to fMRI BOLD signals from depressive vs. non-depressive subjects and identified the Occipital Pole ROI as the strongest-performing region.

Human Activity Recognition — *Scikit-learn, TensorFlow*

- Compared 10 ML models for activity classification across 15 users and 15 activities, including Logistic Regression, Random Forest, XGBoost, and Neural Networks.

TECHNICAL SKILLS

ML/AI: PyTorch, TensorFlow, Scikit-learn, Deep Learning, Computer Vision, Reinforcement Learning, NLP, LLMs
Scientific Computing: PySCF, RT-TDDFT, DFT, Molecular Simulation, QAOA, Qiskit
Data & Analysis: NumPy, Pandas, Matplotlib, Remote Sensing, fMRI Data Analysis, Data Visualization
Programming & Tools: Python, C++, Java, SQL, Git, Linux, Docker, Jupyter

HONORS & ACTIVITIES

Codeforces Rating: 1544 • 1,300+ Problems Solved • ICPC Dhaka Regional Rank: 74th • Graduate Summer Research Fellowship, Miami University (2025) • Dean's List (3×), Rajshahi University